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Network Tomography: Recent Developments

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Domain Signal processing, statistics & applied mathematics

Abstract

Today’s Internet is a massive, distributed network which continues to explode in size as e-commerce and related activities grow. The heterogeneous and largely unregulated structure of the Internet renders tasks such as dynamic routing, optimized service provision, service level verification and detection of anomalous/malicious behavior extremely challenging. The problem is compounded by the fact that one cannot rely on the cooperation of individual servers and routers to aid in the collection of network traffic measurements vital for these tasks. In many ways, network monitoring and inference problems bear a strong resemblance to other “inverse problems” in which key aspects of a system are not directly observable. Familiar signal processing or statistical problems such as tomographic image reconstruction and phylogenetic tree identification have interesting connections to those arising in networking. This article introduces network tomography, a new field which we believe will benefit greatly from the wealth of statistical theory and algorithms. It focuses especially on recent developments in the field including the application of pseudo-likelihood methods and tree estimation formulations.

Real-World Impact & Application

Modern Network Observability (Datadog & ThousandEyes): This seminal IEEE Signal Processing survey synthesized a decade of research across multicast, unicast, loss, and delay tomography into a unified statistical inverse framework. It directly inspired the modern billion-dollar network observability sector, enabling enterprise monitoring platforms like Cisco ThousandEyes, Catchpoint, and Datadog to map complex multi-cloud networks.